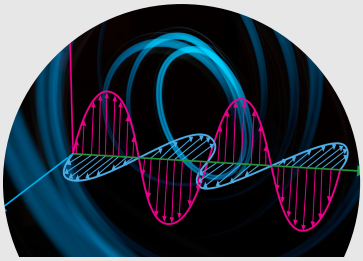

Atmospheric Radiative Transfer

ESCI 345 (3 Credit Hours)

Instructor Info

Dr. Greg Blumberg

Office Hrs: Mon 1:00-2:00 PM, Tues 3:00-5:00 PM, Wed 1:00 PM-3:00 PM

Caputo Hall, Room 409

717-871-7434

http://wblumberg.github.io

Greg.Blumberg@millersville.edu

Course Info

Prereq: C- or higher in ESCI 241, or PHYS 231. Coreq or Prereq: MATH 311.

Tues & Thurs

1:10-2:25 PM

Caputo 400

Course Topics

Quantitative description and analysis of atmospheric radiation and its interaction with atmospheric constituents (gases, aerosol, and clouds) and the land and ocean surfaces. Topics include properties of radiation, the electromagnetic spectrum, reflection and refraction, radiative properties of natural surfaces, thermal emission, atmospheric transmission, atmospheric emission (the Schwarzschild Equation) and absorption, scattering and absorption by molecules and particles, radiative transfer with multiple scattering, numerical modeling of atmospheric radiation, relevance for climate and weather.

Materials

Required Text

A First Course in Atmospheric Radiation, G. Petty, Sundog Publishing

Other Potentially Helpful Texts

Fundamentals of Atmospheric Radiation, C. Bohren and E. Clothiaux
Absorption and Scattering of Light by Small Particles, C. Bohren and D. Huffman

Other required materials (e.g., book chapters, research papers) will be provided on D2L.

Grading Scheme

10%	Participation
40%	Classwork/Homework
25%	Occasional Quizzes & Reflections
25%	Semester Project (TBD)

Grades will follow the standard scale:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
Minimum %	94	90	87	83	80	77	73	70	67	63	60	0

Curving is at the discretion of the professor.

These various assignments are here to help reinforce the material we're learning and let you know how well you have mastered the material. In my mind, a good class has regular questions or comments that contribute to the classroom discussion and demonstrate your thinking. The Classwork/Homework category will consist of various short assignments for you to implement the concepts and skills you've learned in this class, and will be occasionally graded on completion.

There will be no exams in this class, but there will be occasional quizzes or reflections for you to demonstrate what you've learned so far.

Late Work & Absences Policy

Attendance in this class is important as the material will be relevant to your career at some point! You don't want to miss out on a key piece of information that could help you make a better decision! If you need to be absent from class, please let me know ahead of time.

With respect to homework/classwork, I am willing to let you miss one assignment - no questions asked. Occasionally unwelcome life events occur (e.g., health, family issues, etc.). Because I understand how difficult these times can be, I am happy to work with you to see that you are caught up.

Course Learning Objectives

By the end of this course, you should have an understanding of:

- where radiative transfer principles are used in the atmospheric sciences.
- the properties of radiation, such as frequency, wavenumber, and wavelength.
- the difference between broadband and monochromatic radiation.
- the differences between wave and particle representations of electromagnetic radiation and when to use them.
- the relationship between flux and intensity.
- the major spectral bands of the electromagnetic spectrum relevant to the atmospheric sciences.
- the birth, lifespan, and death of photons in the atmosphere.
- how to solve the radiative transfer equation for simplified atmospheric scenarios.
- why certain molecules are prone to atmospheric absorption and emission of radiation.
- how gaseous absorption is modeled in the atmospheric sciences.
- the equations used to characterize thermal emission of radiation and their relationships (e.g., Planck's function, Wien's Displacement law, Stefan-Boltzmann function, Kirchhoff's law).
- representations of atmospheric transmission (extinction, scattering, and absorption coefficients).
- the use of optical depth in understanding the extinction of radiation.
- the calculation of broadband fluxes and heating rates.
- when to incorporate scattering into the radiative transfer equation.
- the methods by which we can include the effects of scattering by particles.

JupyterHub and Python

There will be some Python coding expected in this class. I have worked with Dave Fitzgerald to set up a JupyterHub server for the Earth Sciences department. This is a website that you can log into and access Jupyter notebooks to write Python code. There will be some components of this class where we will be using Python to do some data analysis. The website to access the JupyterHub will be provided in-class.

IMPACTS

This semester I am involved in the IMPACTS field project between January and February. Because I will have to participate in occasional deployments to launch weather balloons (as will others in this class, I imagine), we may have to hold our class virtually. I will send an email out and/or put a message up on D2L to let you know when this may happen.

Contacting Me

I'm invested in your success in this course. Professors like myself hold office hours so students can come by to talk about the course material or ask questions about the field of study we are involved in. If you cannot make it into my office, I also have a virtual meeting room through the Zoom software, which you need to use your university account to access. Please send me an email if you'd like to meet with me virtually!

Here is the link to my Zoom meeting room for Office Hours:

`https://millersville.zoom.us/j/94617438462`

Should you need to contact me over email regarding this class, you must follow the rules below. Doing so will be helpful for me in staying organized and allow me to reply to your request in a timely manner:

1. Put your course number and "ESCI" into your email subject line (e.g., "ESCI 109 - Lab 2 Questions").
2. Use your Millersville account email instead of your personal email.
3. Start and end with a proper greeting (e.g., "Good morning, Dr. Blumberg" "Sincerely, *Your Name*").
4. Check your grammar and punctuation!
5. Be kind and gracious.
6. Be clear about what it is you want to tell me. If you have run into an issue, let me know how you've tried to solve it first!

Diversity and Inclusion Statement

I consider this classroom to be a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class.

Accommodations for Students with Disabilities

If you have a documented physical or learning disability and want to use your special accommodations through the Learning Center, please let me know. Your accommodations may benefit other students in our class as well! Resources for obtaining these accommodations can be found at http://www.millersville.edu/learningservices/disabilityaccom.php.

ADA Program (Office of Learning Services) Americans With Disability Act | Millersville University (if you have a disability that requires accommodations under the Americans with Disabilities Act, please present your letter of accommodations and meet with me as soon as possible so that I can support your success in an informed manner. Accommodations cannot be granted retroactively. If you would like to know more about the Millersville University Office of Learning Services-please contact the office at 717-871-5554)

Millersville University Policies and Links

- Academic Honesty Policy link Governance Manual (millersville.edu); for additional information please see the following: What is Academic Integrity? | Millersville University
- Attendance Policy link: Class Attendance Policy | Millersville University
- Inclusion Statement: Millersville University Inclusion Statement | Millersville University
- Land Acknowledgement: Land Acknowledgement | Millersville University
- Policy on Delays and Cancellations link Policy on Delays & Cancellations | Millersville University
- Preferred Name FAQs link Preferred Name FAQs | Millersville University
- Privacy Rights under FERPA link Annual Notification of Student Rights Under FERPA | Millersville University
- Student Conduct and Community Standards Handbook link studentcodeofconduct.pdf (millersville.edu)
- Title IX Reporting Requirements and the Faculty member: Millersville University is committed to maintaining a safe education environment for all students. In compliance with Title IX of the Education Amendments of 1972 and guidance from the Office for Civil Rights, the University requires faculty members to report incidents of sexual violence shared by students to the University's Title IX Coordinator. The only exceptions to the faculty member's reporting obligation are when incidents of sexual violence are communicated by a student during a classroom discussion, in a writing assignment for a class, or as part of a University-approved research project. Faculty members are obligated to report sexual violence or any other abuse of a students who was, or is, a child (under 18 years of age) when the abuse allegedly occurred to the person. Information about Title IX, resources and reporting can be found at: What is Title IX | Millersville University